

Eric Araújo

Associate Professor

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Welcome

I investigate the spread of opinions, sentiments, and behaviors in complex systems through multiagent modeling and simulations, offering insights into social dynamics in politics, cooperation, health, criminology, and religion.

The work I develop is multidisciplinary, using agent-based models and data science to solve problems in areas such as political polarization in churches, public security, the spread of fake news, and promoting healthier lifestyles.

Education

Ph.D. **Vrije Universiteit Amsterdam**, Computer Science — Amsterdam, NL
- Thesis: Contagious: Modeling the Spread of Behaviours, Perceptions and Emotions in Social Networks

Aug 2014 – Sept 2018

M.Sc. **Universidade Federal de Minas Gerais**, Computer Science — Belo Horizonte, Brazil
Mar 2007 – Sept 2009

B.Sc. **Universidade Federal de Viçosa**, Computer Science — Viçosa, Brazil
Mar 2003 – Dec 2006

Experience

Calvin University, Associate Professor — Grand Rapids MI USA
- Computer Science Department Calvin University

July 2024 – present
2 years 4 months

FIOCRUZ, Researcher — Rio de Janeiro Brazil
- FIOCRUZ is a leading institution in health research and development in Brazil. My work focused on collecting data and analyzing it to understand medical circulation of health sciences among Brazilian researchers.

Jan 2022 – Jan 2024
2 years 1 month

Vrije Universiteit Amsterdam, Guest Researcher — Oct 2018 – Apr 2022
- Collaborator in the Department of Computer Science Vrije Universiteit Amsterdam
- Worked on criminology models with Prof. Dr. Charlotte Gerritsen.

3 years 7 months

Universidade Federal de Lavras, Associate Professor — Minas Gerais Brazil
- Computer Science Department Universidade Federal de Lavras

Feb 2011 – June 2024
13 years 5 months

Gammon Presbyterian Faculty, Professor — Minas Gerais Brazil
- Professor and Coordinator of the Information Systems course

Jan 2010 – Jan 2011
1 year 1 month

Books

Revisiting AI Ethics: A Practice-Grounded Approach to Development, Application, and Policy Making — Feb 2027
Maarten Verkerk, Christine Boshuijzen-van Burken, **Eric Araújo** (Eds.)
link.springer.com/book/9783032412522 (Springer Cham. Forthcoming, February 2027. ISBN 978-3-032-41252-2. Edited volume. Applies Dooyeweerd's modal framework to the ethics of artificial intelligence.)

Publications

Brasil nas redes globais de pesquisa em Ciências da Vida e da Saúde: liderança dependente do contexto (2010–2019) — Sept 2026
Juliana Freitas Lopes, Priscila Costa Albuquerque, **Eric Araújo**, Fabio Zicker, Bruna de Paula Fonseca
10.22477/x.ebbc.1032 (Encontro Brasileiro de Bibliometria e Cientometria (EBBC), v. 10, pp. 1–10 (extended abstract))

Context-dependent leadership in global health science: Brazil's role across North–South, South–South, and triangular collaboration models (2010–2019) — Aug 2026
Juliana Freitas Lopes, Priscila Costa Albuquerque, **Eric Araújo**, Fabio Zicker, Bruna de Paula Fonseca
10.1007/s11192-026-05781-0 (Scientometrics)

Caracterização Temporal de Roubos de Rua Contrastando Contextos Pré-Pandêmico e Pandêmico — June 2026
Renato Figueiredo Frade, **Eric Fernandes de Mello Araújo**, João Paulo Roquim Romanelli, Carlos Henrique da Silveira
10.31060/rbsp.2026.v20.n2.2081 (Revista Brasileira de Segurança Pública)

From brain drain to brain circulation: International researcher mobility in Brazilian health and life sciences (2005–2020) — Feb 2026
Leonardo Biazoli, Priscila Costa Albuquerque, **Eric Araújo**, Izabela Regina Cardoso de Oliveira, Bruna de Paula Fonseca
10.1162/QSS.a.411 (Quantitative Science Studies)

Exploring the link between urban topology and street crime using complex networks: a case study from Southeast Brazil — July 2025
Matheus de Andrade Flausino, **Eric Araújo**, Angélica Sousa da Mata
10.1093/comnet/cnaf016 (Journal of Complex Networks)

The Use of Agent-based Modeling in the Study of Complex Systems (*In Portuguese*) — Jan 2025
Clayton R. da Silva, Olivia Mesquita, **Eric Araújo**, Angélica S. Mata
10.1590/1806-9126-RBEF-2024-0464 (Revista Brasileira de Ensino de Física)

Understanding the complexities of Bluetooth for representing real-life social networks — Jan 2024
Bojan Simoski, Michel C.A. Klein, **Eric Fernandes de Mello Araújo**, Aart T. van Halteren, Thabo J. van Woudenberg, Kirsten E. Bevelander, Moniek Buijzen, Henri Bal
10.1007/s00779-020-01435-x (Personal and Ubiquitous Computing)

Police Planning Based on the Use of Business Intelligence (BI): The Police Manager's Perception Regarding the Violent Crime Rate in the 8th Battalion of the Military Police of Minas Gerais (PMMG) (*In Portuguese*) — Jan 2023
Erich da Costa, Denis Renato de Oliveira, **Eric Araújo**
10.29377/rebesp.v16i01.614 (Revista Brasileira de Estudos de Segurança Pública)

Epidemiological Intelligence in Operation COVID-19: A New Decision Support Tool in Brazil's Military Logistics (*In Portuguese*) Jan 2023
José Roberto Pinho de Andrade Lima, Mariza Ferro, **Eric Araújo**, Cristiano Barros, Ernesto Rademaker Martins, Beatriz Helena Felício Fuck Telles Ferreira
10.47240/revistadaesg.v38i82.1250 (Revista da Escola Superior de Guerra)

Political Polarization on Twitter During the COVID-19 Pandemic: A Case Study in Brazil Jan 2022
Pedro Brum, Matheus Cândido Teixeira, Renato Vimieiro, **Eric Araújo**, Wagner Meira Jr, Gisele Lobo Pappa
10.1007/s13278-022-00949-x (Social Network Analysis and Mining)

Urban Afforestation and Public Safety: A Bibliometric Study Using the Consolidated Meta-Analytical Focus Theory (*In Portuguese*) Jan 2022
Kelly Iapuque Rodrigues de Sousa, Michele Valquíria dos Reis, Rafael Rodrigues de Castro, **Eric Araújo**
10.5585/geas.v11i2.22965 (Revista de Gestão Ambiental e Sustentabilidade (GeAS))

Detecção e Classificação de Bots Utilizando Redes Neurais Artificiais e Análise de Sentimentos Jan 2021
Gabrieli Silva, Eliaquim Ramos, **Eric Araújo**, Fábio Borges, Mariza Ferro
www.eamc.lncc.br/proceedings.html (Proceedings of the XIV Encontro Acadêmico de Modelagem Computacional. ISBN 978-85-99961-27-8)

The Virus and Socioeconomic Inequality: An Agent-based Model to Simulate and Assess the Impact of Interventions to Reduce the Spread of COVID-19 in Rio de Janeiro, Brazil Jan 2020
Vinícius Prata Klôh, Gabrieli Dutra Silva, Mariza Ferro, **Eric Araújo**, Cristiano Barros de Melo, José Roberto Pinho de Andrade Lima, Ernesto Rademaker Martins
10.34119/bjhrv3n2-192 (Brazilian Journal of Health Review)

Disconnecting for the Good: A Network-Oriented Model for Social Contagion of Opinions and Social Network Interventions to Increase Adherence to Social Distancing Jan 2020
Eric Araújo, Mariza Ferro, Gabrieli Silva
10.5753/brasnam.2020.11170 (Proceedings of the IX Brazilian Workshop on Social Network Analysis and Mining)

Creating a Temporal Pattern for Street Robberies Using ABM and Data from a Small City in Southeast Brazil Jan 2020
Eric Araújo, Charlotte Gerritsen
10.4324/9780429277177 (Agent-Based Modelling for Criminological Theory Testing and Development. ISBN 9780429277177)

Identifying Influence Agents That Promote Physical Activity Through the Simulation of Social Network Interventions: Agent-Based Modeling Study Jan 2019
Thabo J. van Woudenberg, Bojan Simoski, **Eric Araújo**, Kirsten E. Bevelander, William J. Burk, Crystal R. Smit, Laura Buijs, Michel Klein, Moniek Buijzen
10.2196/12914 (Journal of Medical Internet Research)

A Social Network Model for Integration of Refugees Jan 2019
Fabio Curi, Dimitris Nikolopoulos, **Eric Araújo**
10.5220/0007930601650175 (Proceedings of the 9th International Conference on Simulation and Modeling Methodologies, Technologies and Applications. ISBN 9789897583810)

Parameter Optimization for Deriving Bluetooth-Based Social Network Graphs Jan 2019
Bojan Simoski, Michel C.A. Klein, **Eric Araújo**, Aart T. Van Halteren, Thabo Van Woudenberg, Kirsten E. Bevelander, Moniek Buijzen, Henri Bal
10.1109/SmartWorld-UIC-ATC-SCALCOM-IOP-SCI.2019.00318 (2019 IEEE SmartWorld, Ubiquitous Intelligence & Computing, Advanced & Trusted Computing, Scalable Computing & Communications, Cloud & Big Data Computing, Internet of People and Smart City Innovation (SmartWorld/SCALCOM/UIC/ATC/CBDCom/IOP/SCI))

Contagious: Modeling the Spread of Behaviours, Perceptions and Emotions in Social Networks (PhD Thesis) Jan 2018
Eric Araújo
research.vu.nl/en/publications/632878cc-e2e7-40ab-a34e-91bde40eaf50 (Vrije Universiteit Amsterdam. ISBN 9789402811179)

Using Simulations for Exploring Interventions in Social Networks - Modeling Physical Activity Behaviour in Dutch School Classes Jan 2018
Eric Araújo, Bojan Simoski, Thabo van Woudenberg, Kirsten Bevelander, Crystal Smit, Laura Buijs, Michel Klein, Moniek Buijzen
10.5220/0006857704140425 (Proceedings of 8th International Conference on Simulation and Modeling Methodologies, Technologies and Applications - SIMULTECH. ISBN 978-989-758-323-0)

Applying Machine Learning Algorithms for Deriving Personality Traits in Social Network Jan 2018
Eric Araújo, Bojan Simoski, Michel Klein
10.1145/3167132.3167377 (Proceedings of the 33rd Annual ACM Symposium on Applied Computing. ISBN 9781450351911)

Detecting Dutch Political Tweets: A Classifier based on Voting System using Supervised Learning Jan 2018
Eric Araújo, Dave Ebbelaar
10.5220/0006592004620469 (Proceedings of the 10th International Conference on Agents and Artificial Intelligence - Volume 1: ICAART. ISBN 978-989-758-275-2)

Social Connection Dynamics in a Health Promotion Network Jan 2017
Eric Araújo, Michel Klein, Aart van Halteren
10.1007/978-3-319-50901-3_61 (Complex Networks & Their Applications V. ISBN 978-3-319-50901-3)

Explaining Changes in Physical Activity Through a Computational Model of Social Contagion Jan 2017
Julia S. Mollee, **Eric Araújo**, Adnan Manzoor, Aart T. van Halteren, Michel Klein
10.1007/978-3-319-54241-6_19 (Complex Networks VIII. ISBN 978-3-319-54241-6)

Exploring Parameter Tuning for Analysis and Optimization of a Computational Model Jan 2017
Julia S. Mollee, **Eric Araújo**, Michel C. A. Klein
10.1007/978-3-319-60045-1_36 (Advances in Artificial Intelligence: From Theory to Practice. ISBN 978-3-319-60045-1)

A Temporal-Causal Model for Spread of Messages in Disasters Jan 2017
Eric Araújo, Annelore Franke, Rukshar Wagid Hosain
10.1007/978-3-319-67077-5_37 (Computational Collective Intelligence. ISBN 978-3-319-67077-5)

A Computational Cognitive Model for Political Positioning and Reactions in Web Media Jan 2017
Eric Araújo, Michel Klein
10.1109/ICCI-CC.2017.8109782 (2017 IEEE 16th International Conference on Cognitive Informatics & Cognitive Computing (ICCI*CC))

Analysis and Refinement of a Temporal-Causal Network Model for Absorption of Emotions Jan 2016
Eric Araújo, Jan Treur
10.1007/978-3-319-45243-2_3 (Computational Collective Intelligence. ISBN 978-3-319-45243-2)

Online Sharing of Physical Activity: Does It Accelerate the Impact of a Health Promotion Program? Jan 2016
Adnan Manzoor, Julia S. Mollee, **Eric Araújo**, Aart T. van Halteren, Michel C. A. Klein
10.1109/BDCloud-SocialCom-SustainCom.2016.40 (2016 IEEE International Conferences on Big Data and Cloud Computing (BDCloud), Social Computing and Networking (SocialCom), Sustainable Computing and Communications (SustainCom) (BDCloud-SocialCom-SustainCom))

Analysis and Evaluation of Social Contagion of Physical Activity in a Group of Young Adults Jan 2015
Eric Araújo, Anita V. T. T. Tran, Julia S. Mollee, Michel C. A. Klein
10.1145/2818869.2818922 (Proceedings of the ASE BigData & SocialInformatics 2015. ISBN 9781450337359)

An Analysis of TCP Protocol Issues in Asymmetric Networks and Existing Solutions (*In Portuguese*) - M.Sc. Thesis Jan 2009
Eric Araújo
hdl.handle.net/1843/BUBD-9JWQ36 (Universidade Federal de Minas Gerais)

AnimEasy: Free Tool for Digital Inclusion Focused on Multimedia Using Inkscape (*In Portuguese*) - B.Sc. Senior Project Jan 2007
Eric Araújo, Ricardo dos Santos Ferreira
(VII Simpósio de Informática do Planalto Médio (SIPM 2007))

Events

Wisdom in the Age of AI Conference — Church track Calvin University, Grand Rapids, MI Oct 2026
"Modeling Communities in Peril: Polarization and the Changes in the Social Fabric of Congregations through an Agent-Based Approach." Read paper on the BBB (Belief, Behavior, Belonging) agent-based model of congregational polarization, with the Brazilian and US context for church fragmentation.

CompCOVID 2020 Brazil (online) June 2020
Panel on computing applications during the COVID-19 pandemic. In Portuguese.

IX Brazilian Workshop on Social Network Analysis and Mining (BraSNAM 2020) Brazil (online) Dec 2020
Conference presentation of BraSNAM 2020 paper (doi:10.5753/brasnam.2020.11170).

Saúde Única — UFLA Lavras, Brazil (online) Feb 2021
Talk on opinion contagion in social networks and its epidemiological impact. In Portuguese.

ABC²-Tec 5ª Reunião — Associação Brasileira de Cristãos na Ciência Brazil (online) July 2021
Talk on social contagion models at the Brazilian Association of Christians in Science. In Portuguese.

KDMiLe 2021 — Symposium on Knowledge Discovery, Mining and Learning — Brazil (online)
Keynote session (KSN2) at KDMiLe 2021. In Portuguese. — Oct 2021

PROCORES Research Group — Brazil (online)
Research group talk on social network spreading models. In Portuguese. — Dec 2022

Oficina Prograd — UFLA — Lavras, Brazil
Workshop on AI tools for undergraduate education. In Portuguese. — Mar 2024

Departamento de Estatística — UFLA — Lavras, Brazil
Invited talk on data-driven agent-based modeling. In Portuguese. — Oct 2024

19º Congresso Brasileiro de Sistemas (CBS) — Goiânia, Brazil
In Portuguese — Nov 2024

Autonomous Agents for Social Good (AASG) 2025 — Detroit, MI
Presented by students Daniel Kwon and Jaden Brookens (senior project) — May 2025

Algoritmos e Sociedade — UFG — Goiânia, Brazil
In Portuguese — May 2025

INSNA Sunbelt 2025 — Paris, France
With Juliana Freitas Lopes et al. (Fiocruz) — June 2025

American Scientific Affiliation (ASA) Annual Meeting 2025 — Colorado Christian University, Lakewood, CO
July 2025

Grace Hopper Celebration 2025 — Chicago IL USA
The Grace Hopper Celebration is the world's largest gathering of women in computing. Attended with the Calvin Girls Who Code student delegation. — Nov 2025

Baylor Symposium on Faith & Culture 2026 — Technology and the Human Person in the Age of AI — Waco, TX
Feb 2026
"Cognitive Modeling of Christian Communities: Developing Intelligent Agents to Address the Polarization Effects in Churches." Session "AI and Relationships in Workplace Communities" (chair: Tiffany Flood), Baines, Bill Daniel Student Center, February 26, 2026, 12:00–1:30 p.m.

NetLogo Conference 2026 — Chicago, IL
Presented by students Katelin Jandris, Ryan Klein, and Övgü Tufan (senior project) — July 2026

Prompted: Liberal Arts in the Age of AI — Season 2, Episode 6 — Calvin University (podcast)
Podcast guest, discussing the Matristics summer research on what leading large language models know about two fourth-century Christian women, and previewing the October 8–10 "Wisdom in the Age of AI" conference. Hosted by Craig Mattson; production funded by the Calvin Center for Christian Scholarship. — Aug 2026

American Academy of Religion (AAR) Annual Meeting 2026 — History of Christianity Unit — Denver, CO
Papers session "History of the Present, the Futures of Christianity's Past" (A23-217), Monday Nov 23, 12:30–2:30 PM. Empirical evaluation of LLM accuracy on Macrina the Younger and Olympias of Constantinople. — Nov 2026

Grants and Fellowships

Templeton World Charity Foundation — Planning Grant June 2026 – Jan 2027
$100,000 planning grant for a research program on polarization and listening across communities, awarded under the Templeton World Charity Foundation's Listening and Learning in a Polarized World (LLPW) program and co-led with Dr. Damaris Parsitau. Extends the BBB (Belief, Behavior, Belonging) agent-based modeling work to congregations in Brazil and Kenya, with ethnographic fieldwork preceding the computational modeling. Full implementation proposal due January 15, 2027.

Marjorie G. Hoogeboom Student Faculty Research Fellowship Summer 2026
Inaugural award of the Hoogeboom fellowship, supporting a 10-week summer research placement for student David Quinn on the empirical validation track of the TWCF church-polarization project: survey-data pipelines for initializing agent belief distributions, calibration targets drawn from the social science literature, and support for the international pilot survey instruments.

Calvin Research Fellowship 2026 – 2027
Awarded 4 faculty load hours for the 2026–2027 academic year for the project "Modeling Political and Religious Polarization in Christian Communities." Named a Diekema Fellow, with support paid through the Diekema Endowment.

Nagel Institute Fellowship Jan 2026
Supporting research on computational modeling of world Christian communities.

McGregor Summer Research Fellowship May 2026 – Aug 2026
Supporting collaborative faculty-student research.

Teaching

Introduction to Data Structures Calvin University USA
An introduction to data structures and algorithms, focusing on the design and analysis of algorithms, data structures, and their applications. Sept 2024 – present

Databases Management Systems Calvin University USA
An introduction to database management systems, covering data modeling, query languages, and database design. Jan 2025 – present

Agent-based Modeling Calvin University USA
An introduction to agent-based modeling, focusing on the design and implementation of agent-based models for complex systems. Sept 2024 – present

Concepts of Programming Languages Calvin University USA
Concepts and paradigms of programming languages. Jan 2026 – present

Research Projects

Matristics: Evaluating Large Language Models on the Women of the Early Church Calvin University
With D. Rylaarsdam. Supported by a McGregor Undergraduate Research Fellowship. 2026 – present

- A public benchmark testing whether leading large language models (Claude, ChatGPT, Gemini, Grok) reproduce the androcentric silences of the historical record about women in early Christianity.
- First paper accepted to the American Academy of Religion Annual Meeting 2026.

Tolle Lege — Pocket Classics to Print and Sew Calvin University
Founder, editor and typesetter. 2026 – present

- Free pocket-size booklets of short public-domain classics on attention, study and the spiritual life, each in the most recent public-domain translation, laid out for home printing and hand binding. Begun for a student reading group at Calvin University; layouts CC BY 4.0, code MIT.

CS Prayers — A Book of Hours for Computing Calvin University
Founder and curator. 2026 – present

- A public collection of liturgies for the rhythms of studying, teaching, and practicing computer science, open to contributions from students and faculty at any institution.

Procores UFMG Brazil
Research group funded by CNPq (Brazil) May 2022 – present

- ProCoReS is a research group funded by CNPq, dedicated to the study of social networks, mathematical modeling, contagion processes, and the analysis of complex data. The group brings together researchers from different fields to propose and analyze innovative models that consider multiple aspects of social networks and their dynamics.
- To model networks and contagion, most current approaches use graphs. However, these are not entirely realistic, and new models of contagion processes should allow for: representation of different classes of individuals and connections, linkage between concurrent and co-occurring processes, and connections involving two or more individuals simultaneously. Understanding how connections are formed and how different types of flows permeate such networks is crucial to understanding how our society is organized and evolves. The goal is to propose mathematical, statistical, agent-based, and machine learning models that account for: multiple complex aspects of social contagion processes and parameterization through data sources (e.g., online social networks, mobility, collaborations). Such models should allow for classes of individuals and relationships; model co-occurrence, causality, and links between contagion processes; and represent multi-relationships.
- Funded by: CNPq Universal 2022

Socioeconomic Impacts in Territories Affected by Technological Disasters UFLA Brazil
Sept 2022 – Feb 2024

- This project investigates the socioeconomic consequences of technological disasters in Brazil, which have increased significantly in recent decades. Between 2000 and 2020, Brazil recorded 96 technological disasters—ranging from dam collapses to oil spills—impacting over 42 million people and causing over USD 17 billion in damages (EM-DAT, 2021). Unlike natural disasters, which occur without human interference, technological disasters result from anthropogenic actions, such as industrial accidents or environmental contamination. These events often lead to material loss, environmental degradation, and serious public health issues, demanding robust mechanisms for community response and risk management.
- By analyzing historical data and specific case studies such as the Mariana (2015) and Brumadinho (2019) dam collapses, the project seeks to understand how these disasters affect the social and economic fabric of impacted territories. The ultimate goal is to identify patterns and develop frameworks that can inform policies for disaster prevention, mitigation, and recovery—contributing to more resilient communities in the face of growing technological risks.

Analysis of User Behavior on a University Campus Wi-Fi Network UFLA Brazil

Masters Project by Thiago do Prado Ramos Mar 2021 – Sept 2023

- This project aims to analyze the behavior of users on a university campus Wi-Fi network, focusing on connection patterns and user mobility. By examining connection logs, the project seeks to understand how users interact with the network, their mobility patterns, and how these factors influence network performance and user experience.
- The findings will help in optimizing network infrastructure, improving user experience, and informing future network design and management strategies.

Algorithms for the Optimization of Supply Chain Management Systems UFLA Brazil

Project in partnership with Prof. Mayron Moreira (UFLA Brazil) Mar 2021 – Mar 2022

- This project developed and evaluated algorithms aimed at optimizing key components of supply chain management, including inventory control, logistics, and production scheduling. By modeling supply chains as dynamic systems, the project applied heuristic and metaheuristic approaches—such as genetic algorithms and simulated annealing—to improve efficiency, reduce costs, and enhance decision-making under uncertainty.
- The research focused on real-world constraints such as demand variability, delivery time windows, and resource limitations. The resulting algorithms demonstrated significant improvements in performance when compared to traditional methods, and provided a flexible framework that can be adapted to various industrial sectors seeking to enhance their operational processes.

Urban Tree Cover and Crime UFLA Brazil

Project in partnership with Prof. Michele Valquíria dos Reis (UFLA Brazil) Mar 2020 – Mar 2024

- Urban tree cover provides vital ecosystem services that promote public health and well-being. However, there is a common perception—found in media reports and informal discussions—that trees may facilitate crime by obstructing visibility or serving as hiding places. While some studies have linked vegetation to fear of crime, often recommending vegetation removal to improve safety, research from the Northern Hemisphere increasingly suggests a negative correlation between urban greenery and crime rates, indicating that vegetation may actually contribute to crime reduction.
- This project aims to investigate the relationship between urban tree cover and crime incidence in Brazilian cities. It integrates data from street tree inventories, georeferenced crime records from the Military Police, socioeconomic indicators from the national census (IBGE), citizen surveys on perceptions of vegetation and safety, and citizen requests for tree removal. The goal is to generate scientific insights into the social dimensions of urban greenery and to inform evidence-based environmental management policies. The findings are intended to support public agencies and civil society in designing safer, greener urban spaces.

Computational Modeling and Simulation of Contagion in Social Networks UFLA Brazil

Sept 2018 – June 2024

- Social contagion refers to the spread of behaviors, perceptions, and emotions through interactions among individuals. Often occurring unconsciously, this phenomenon plays a role in shaping collective opinion, promoting positive behaviors, and influencing public discourse. This project aims to simulate the diffusion of information originating from social media platforms across various contexts, using agent-based and network modeling approaches.
- The study investigates how network topology, individual agent characteristics, and intervention strategies affect the spread and quality of information. It explores methods for enhancing the visibility and influence of high-quality content while limiting the reach of misinformation or fake news. The results aim to inform the design of healthier and more reliable information ecosystems, with potential applications in communication policy, platform design, and digital education.

Supervised Students

Miriam Crisman — Calvin University
B.S. - Computational Modeling of Political Polarization in Complex Systems — Fall 2026 - Spring 2027
- Senior Project

Adam Byle, Alayna Smith — Calvin University
B.S. - Data-Oriented Agent-Based Model of Polarization and Fracture in Kenyan Society — Fall 2026 - Spring 2027
- Senior Project

Sally Amonoo-Mensah — Calvin University
B.S. - What Is AI Saying about Junia? — Fall 2026 - Spring 2027
- Senior Project

David Quinn — Calvin University
B.S. - Empirical Validation and Data Pipelines for an Agent-Based Model of Church Polarization — Summer 2026
- Marjorie G. Hoogeboom Student Faculty Research Fellowship

Miriam Crisman — Calvin University
B.S. - NetLogo Agent-Based Model of Church Community Polarization — Summer 2026
- Part of the Templeton World Charity Foundation planning grant (Listening and Learning in a Polarized World)

Brady Swiech, Tim David, Zach Redder — Calvin University
B.S. - Scale66 — AI Platform for Marketing Content Generation — Fall 2025 - Spring 2026

Esther Asuquo, David Barry — Calvin University
B.S. - Modeling Gram-Negative Bacterial Infections: An Agent-Based Approach — Spring 2025 - Fall 2025

Katelin Jandris, Ryan Klein, Övgü Tufan — Calvin University
B.S. - Cognitive Modeling of Church Polarization — Fall 2025 - Spring 2026
- Presented at NetLogo Conference 2026 (Chicago — July 2026)

Jaden Brookens, Daniel Kwon — Calvin University
B.S. - Modeling Introversion in the Classroom: An Agent-Based Approach — Fall 2024 - Spring 2025
- Conference paper at AASG 2025 (Detroit)

Leonardo Biazoli — Federal University of Lavras (UFLA) — Brazil
Ph.D. - Large-Scale Bibliometric Analysis of International Researcher Mobility in Brazilian Health and Life Sciences — 2022 - 2025
- Published in Quantitative Science Studies (2026)
- DOI: 10.1162/QSS.a.411

Clayton Ramos da Silva — Federal University of Lavras (UFLA) — Brazil
M.S. - Leishmaniasis Transmission Modeling Using Agent-Based Models and Complex Networks — 2022 - 2024
- Published in Revista Brasileira de Ensino de Física (2025)
- DOI: 10.1590/1806-9126-RBEF-2024-0464

Matheus de Andrade Flausino — Federal University of Lavras (UFLA) — Brazil
M.S. - Urban Topology and Street Crime Patterns Using Complex Networks — 2022 - 2024
- Published in Journal of Complex Networks (2025)
- DOI: 10.1093/comnet/cnaf016

Thiago Guedes de Jesus — Federal University of Lavras (UFLA) — Brazil
M.S. - Epidemiological Modeling with Evolutionary Game Theory in Spatial Networks — 2022 - 2024

Rolf Pagotto Veiga — Federal University of Lavras (UFLA) — Brazil
M.S. - Diretrizes para Classificação Automática de Músicas Cristãs: Aplicação do BERT na Identificação de Adoração — 2021 - 2024
- M.S. defended August 30, 2024 — PPGCC/UFLA

Alessandra Louzada Terra — Federal University of Lavras (UFLA) — Brazil
M.S. - Aprendizagem de Máquina para Estudo de Dados em Grafos: Um Estudo do Movimento dos Estudantes de Medicina no Brasil — 2024 - 2025
- M.S. defended January 30, 2025 — PPGCC/UFLA

Kelly Iapuque Rodrigues de Sousa — Federal University of Lavras (UFLA) — Brazil
M.S. - Urban Afforestation and Public Safety: A Bibliometric Study — 2021 - 2023
- Published in Revista de Gestão Ambiental e Sustentabilidade (GeAS) (2022)
- DOI: 10.5585/geas.v11i2.22965

Thiago do Prado Ramos — Federal University of Lavras (UFLA) — Brazil
M.S. - Wireless Network Capacity Planning from Campus Connection Logs — 2021 - 2023

Eduardo Vinicius Betim — Universidade Federal de Santa Catarina (UFSC) — Brazil
B.S. - Análise do Comportamento de Comunidade sobre Dados da Guerra da Ucrânia no Reddit — Jan 2024

Adilson Henrique de Oliveira Lima — Federal University of Lavras (UFLA) — Brazil
B.S. - Modelagem baseada em agentes para estudos sobre transmissão de HIV e comportamento voltado para a prevenção da disseminação do vírus — 2022 - 2023
- Project interrupted (2023)

Débora Rossini Martins Cardoso — Federal University of Lavras (UFLA) — Brazil
B.S. - Impactos do Ensino Remoto durante a Pandemia no Ensino Superior Brasileiro — Jan 2022

Eduardo Miranda Pedrosa Filho — Federal University of Lavras (UFLA) — Brazil
B.S. - Estudo e Implementação de Melhorias em um Sistema de Streaming para Vídeos de Educação Cristã — Jan 2021

Erich Fernando da Costa — Federal University of Lavras (UFLA) — Brazil
B.S. - Planejamento Policial a partir do Uso de Ferramentas de Business Intelligence: Análise do Índice de Criminalidade Violenta nos Municípios do 8º BPM — Jan 2020

Héuller Soares Vilela Silva — Federal University of Lavras (UFLA) — Brazil
B.S. - Previsão do Resultado Final de Partidas de Futebol a partir do Primeiro Tempo usando Técnicas de Aprendizagem de Máquina — Jan 2020

Ana Carolina Le Senechal — Federal University of Lavras (UFLA) — Brazil
B.S. - Análise e Pré-processamento de Dados Utilizando Técnicas de Mineração de Dados para o Moodle — Jan 2013

Giulia Luan Santos de Sousa — Federal University of Lavras (UFLA) — Brazil
B.S. - Tweetmining: Análise de Opinião Contida em Textos Extraídos do Twitter — Jan 2012

Larissa Avelino Zuppo — Federal University of Lavras (UFLA) — Brazil
B.S. - Webometria e Análise de Impacto da Informação em Redes Sociais — Jan 2012

Paulo Roberto de Almeida — Federal University of Lavras (UFLA) — Brazil
B.S. - Validação do Modelo Matemático para a Análise do Comportamento do Protocolo TCP em Redes Assimétricas — Jan 2011

Karen Aline Alves Bezerra — Federal University of Lavras (UFLA) — Brazil
B.S. - BDD: Avaliação e Proposta de Aplicação no Pós Desenvolvimento — Jan 2011

Arthur Moreira Maia — Federal University of Lavras (UFLA) — Brazil
B.S. - Modelagem da Estrutura e Dinâmicas de Redes Sociais: Um Modelo Formal para Estudos de Fenômenos Sociodinâmicos — Jan 2020
- Undergraduate research (Iniciação Científica) — CNPq

Aline Rodrigues Guimarães de Oliveira — Federal University of Lavras (UFLA) — Brazil
B.S. - Modelagem e Simulação de Difusão de Informação em Redes Sociais — Jan 2019
- Undergraduate research (Iniciação Científica) — CNPq

Álvaro Martins Espíndola — Federal University of Lavras (UFLA) — Brazil
B.S. - Uso de Modelagem Baseada em Agentes para Redução de Oportunidade para Agentes Criminosos na Cidade de Lavras — Jan 2019
- Undergraduate research (Iniciação Científica)

Leonardo Amorim de Sena — Federal University of Lavras (UFLA) — Brazil
B.S. - Desenvolvimento e Análise de Plataforma para Estudos sobre Contágio em Redes Sociais — Jan 2019
- Undergraduate research (Iniciação Científica)

Arthur Moreira Maia — Federal University of Lavras (UFLA) — Brazil
B.S. - Modelagem e Simulação de Difusão de Informação em Redes Sociais — Jan 2019
- Undergraduate research (Iniciação Científica) — CNPq

Danielle Aparecida Alcântara — Federal University of Lavras (UFLA) — Brazil
B.S. - Webometria e Análise de Valor na Web — Jan 2013
- Undergraduate research (Iniciação Científica)

Ana Carolina Le Senechal — Federal University of Lavras (UFLA) — Brazil
B.S. - Validação do Modelo Matemático para a Análise do Comportamento do Protocolo TCP em Redes Assimétricas — Jan 2011
- Undergraduate research (Iniciação Científica)

Yasmin Ribeiro Penha — Federal University of Lavras (UFLA) — Brazil
B.S. - Estudo de Algoritmos para Mineração de Opinião para Detecção das Percepções sobre a Cidade de Lavras — Jan 2011
- Undergraduate research (Iniciação Científica)

Ian Hawthorne, Nathan Holtrop — Calvin University
B.S. - Teaching Ancient Wisdom with Modern AI: Evaluating LLMs in Early Christian Studies — Summer 2026
- McGregor Undergraduate Research Fellowship

Caleb Ausema — Calvin University
B.S. - Agent-Based Modeling for Archaeology — Spring 2026 - Fall 2026
- Senior Project

Martin Liu — Calvin University
B.S. - VerseNet: A Visual Neo4j-Based Bible Knowledge Network — Spring 2026 - Fall 2026
- Senior Project

University Service

Track Leader, AI & STEM - Wisdom in the Age of AI Conference, Calvin University, Center for Faith and Technology — Oct 2026
Led the AI & STEM track of the Wisdom in the Age of AI conference (October 8–10, 2026, more than 600 participants), at which Calvin announced its Center for Faith and Technology.

Promotions Committee, Computer Science, Calvin University — 2026 – present
Responsible for the department's Promotions Committee: publicizing the Computer Science program and shaping recruitment strategy.

Planning team, School of Engineering and Computer Science, Calvin University — 2026 – present
Member of the team shaping Calvin's new School of Engineering and Computer Science.

Faculty planning group, CCCS grant on political polarization, Calvin University — 2025 – present
Member of a faculty group planning a Calvin Center for Christian Scholarship (CCCS) grant on political polarization.

Interviewer, prospective Honors students, Calvin University — 2025 – present
Interviews prospective students for the Calvin Honors Program.

Faculty Advisor, Girls Who Code, Calvin University — 2025 – present
Faculty advisor to the Girls Who Code chapter at Calvin University, including leading the annual student delegation to the Grace Hopper Celebration.

Professional Service

Faculty Fellow, Nagel Institute for the Study of World Christianity — Dec 2025

External Member - Masters Exam — Stephano Daniel Santos, UFLA Brazil — Aug 2025

External Member - PhD Qualifying Exam — Jamile Santos dos Santos, UFRJ Brazil — Aug 2024

External Member - Qualifying Exam — Rolf Pagotto Veiga, UFLA Brazil — Aug 2024

Church and Community Service

Elder - Plymouth Heights Christian Reformed Church, Grand Rapids, MI — May 2025 – present
Assists with liturgy; hospitality ministry to college students.

Professional Development

CCCS Writing Co-Op — Calvin University
Week-long writing residency: "Temperance and the Digital Life" (article for The Banner). — July 2026

G&M Young Leaders Training Conference — The Grace & Mercy Foundation — New York, NY — July 2026
Participated in the Grace & Mercy Foundation Young Leaders Training Conference in New York (July 7–12, 2026), on implementing the Public Reading of Scripture and the Just Show Up book club at Calvin University. Trip funded by the Grace & Mercy Foundation and Calvin Theological Seminary.

Faith & Learning Fellowship — de Vries Institute for Global Faculty Development, Calvin University — Sept 2025 – Dec 2025
Faculty fellowship of the de Vries Institute, on integrating Christian faith with scholarly work.

Healthy Dialogue Series — Jennifer Powell McNutt — Calvin University — Feb 2025
Faculty session with Jennifer Powell McNutt (February 18, 2025).

CCCS Writing Co-Op — Calvin University — Jan 2025
Week-long writing residency: "Prudence and the Digital Life."

Healthy Dialogue Series — Fellipe do Vale — Calvin University — Nov 2024
Faculty/staff book discussion with author Fellipe do Vale (November 6, 2024).

Kuiper Seminar — Calvin University — Sept 2024 – May 2025
Year-long seminar on faith and learning integration, pedagogy, and the Reformed tradition.

Research Skills

Agent-based modeling: Netlogo and Python tools for ABMs.

Data Science and Machine Learning: Python, R, Statistics.

Social Network Analysis: Gephi, NetworkX.

Network-oriented modeling: Techniques for modeling complex systems using network-oriented methods.

Programming: Proficient with Python, C++, Netlogo, and Git; good understanding of Web development.

Languages: English (fluent), Portuguese (native), Dutch (basic).